(a) Test utility

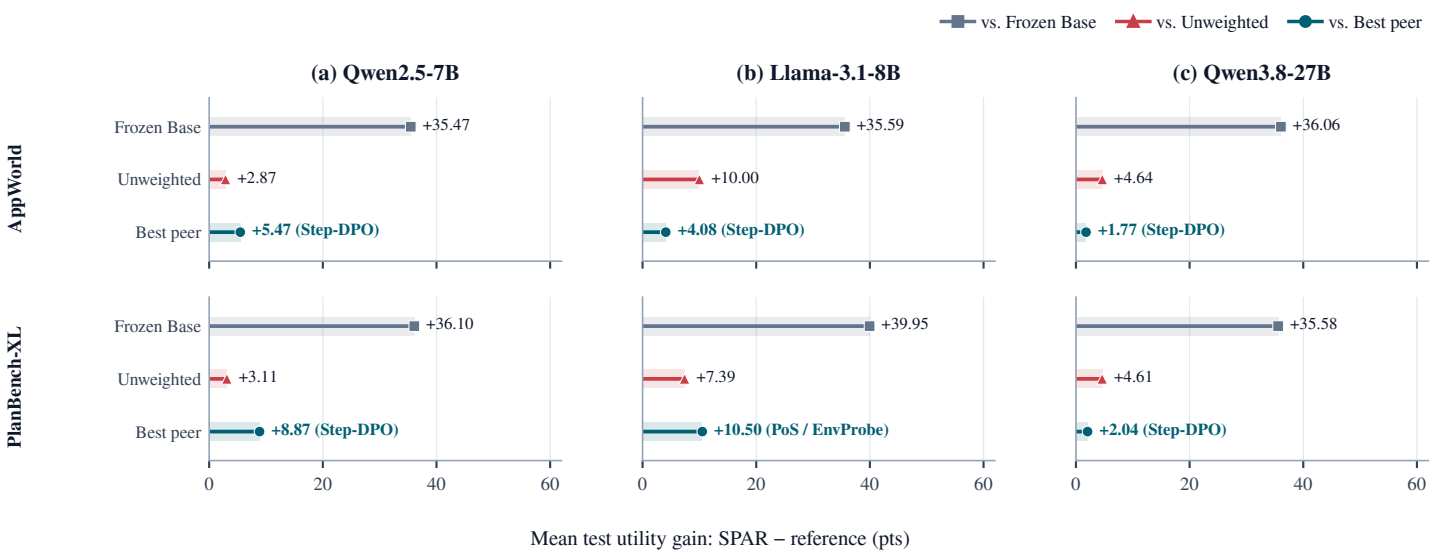
Mean ± sample SD

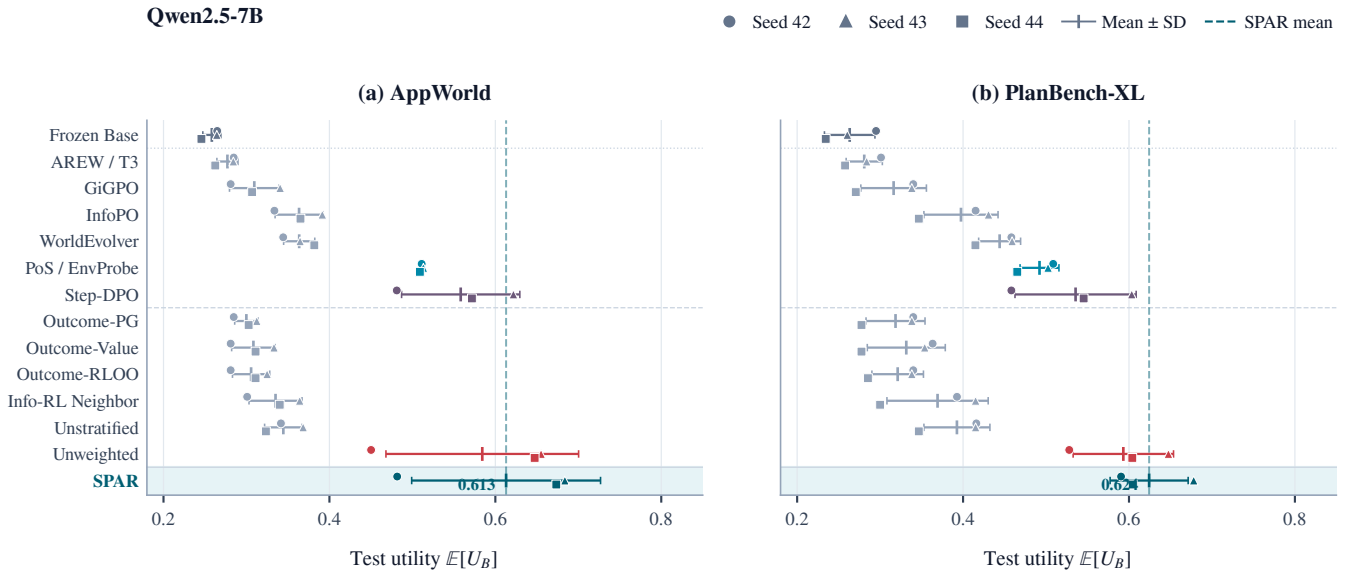
Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL
Frozen Base	0.2583 ±0.0109	0.2633 ±0.0304	0.2720 ±0.0121	0.2682 ±0.0338	0.3341 ±0.0101	0.3290 ±0.0429
AREW / T3	0.2771 ±0.0127	0.2809 ±0.0218	0.2826 ±0.0087	0.3068 ±0.0433	0.3665 ±0.0085	0.3744 ±0.0377
GiGPO	0.3094 ±0.0299	0.3163 ±0.0395	0.3295 ±0.0268	0.3419 ±0.0558	0.3952 ±0.0071	0.4070 ±0.0395
InfoPO	0.3634 ±0.0289	0.3975 ±0.0447	0.3955 ±0.0035	0.4163 ±0.0449	0.5268 ±0.0676	0.5279 ±0.0689
WorldEvolver	0.3636 ±0.0188	0.4441 ±0.0253	0.3827 ±0.0052	0.4541 ±0.0347	0.4487 ±0.0081	0.5227 ±0.0228
PoS / EnvProbe	0.5112 ±0.0021	0.4921 ±0.0234	0.5583 ±0.0059	0.5627 ±0.0224	0.6649 ±0.0104	0.6630 ±0.0081
Step-DPO	0.5583 ±0.0713	0.5356 ±0.0730	**0.5871** ±0.0636	0.5607 ±0.0658	**0.6770** ±0.0398	**0.6643** ±0.0358
Outcome-PG	0.2999 ±0.0141	0.3186 ±0.0356	0.3138 ±0.0147	0.3520 ±0.0704	0.4078 ±0.0332	0.4099 ±0.0353
Outcome-Value	0.3084 ±0.0262	0.3316 ±0.0470	0.3310 ±0.0219	0.3693 ±0.0601	0.4104 ±0.0302	0.4099 ±0.0353
Outcome-RLOO	0.3056 ±0.0224	0.3212 ±0.0311	0.3188 ±0.0266	0.3546 ±0.0660	0.4038 ±0.0384	0.4099 ±0.0353
Info-RL Neighbor	0.3350 ±0.0320	0.3693 ±0.0610	0.3633 ±0.0197	0.3900 ±0.0511	0.4635 ±0.0390	0.4233 ±0.0412
Unstratified	0.3444 ±0.0226	0.3926 ±0.0398	0.3808 ±0.0100	0.3998 ±0.0399	0.4712 ±0.0137	0.4743 ±0.0397
Unweighted	**0.5842** ±0.1161	**0.5932** ±0.0605	0.5279 ±0.1105	**0.5939** ±0.0828	0.6484 ±0.0432	0.6386 ±0.0444
SPAR	**0.6130** ±0.1138	**0.6243** ±0.0470	**0.6279** ±0.1102	**0.6677** ±0.0197	**0.6948** ±0.0091	**0.6847** ±0.0130

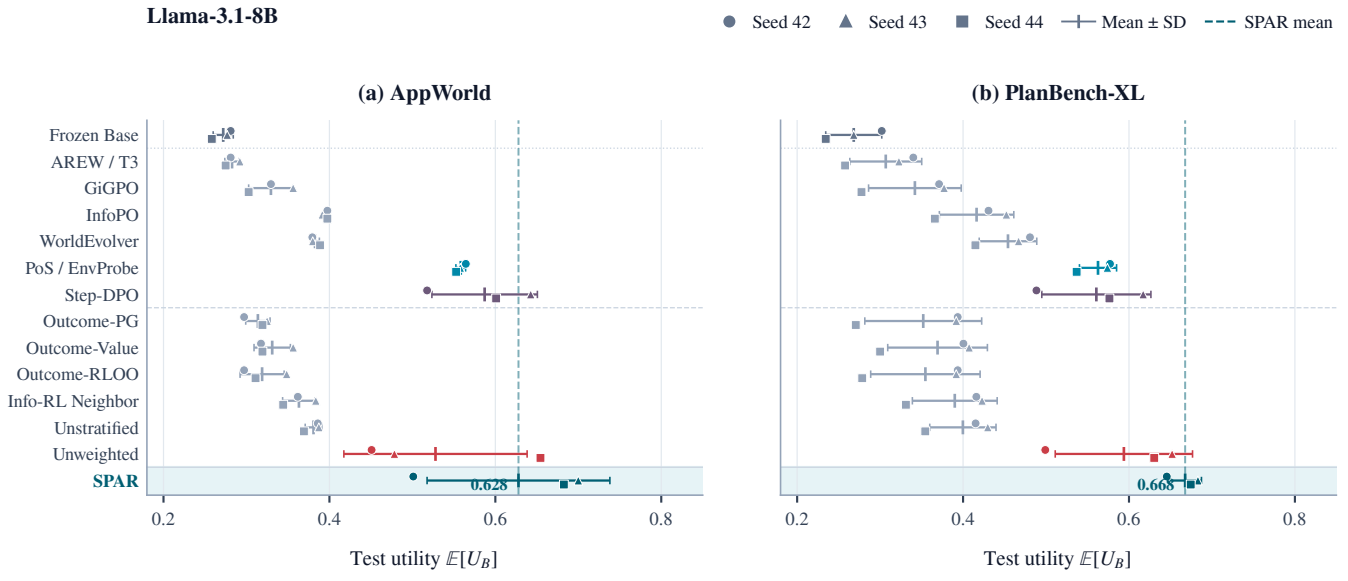
(b) Test success (%)

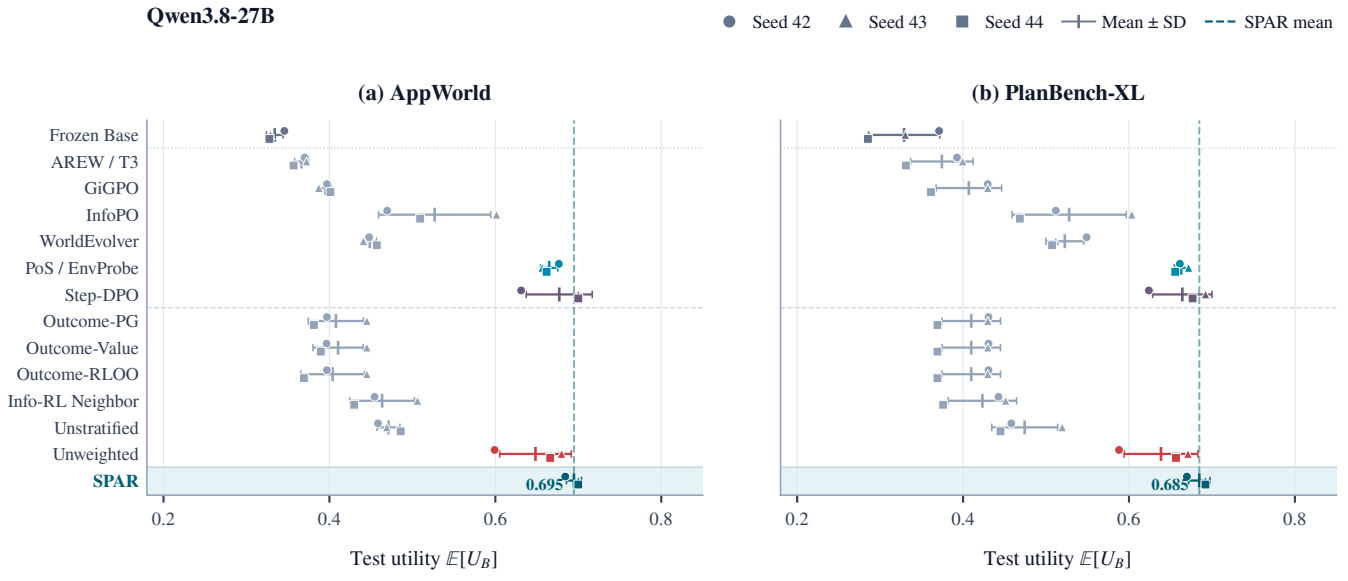
Mean ± sample SD (%)

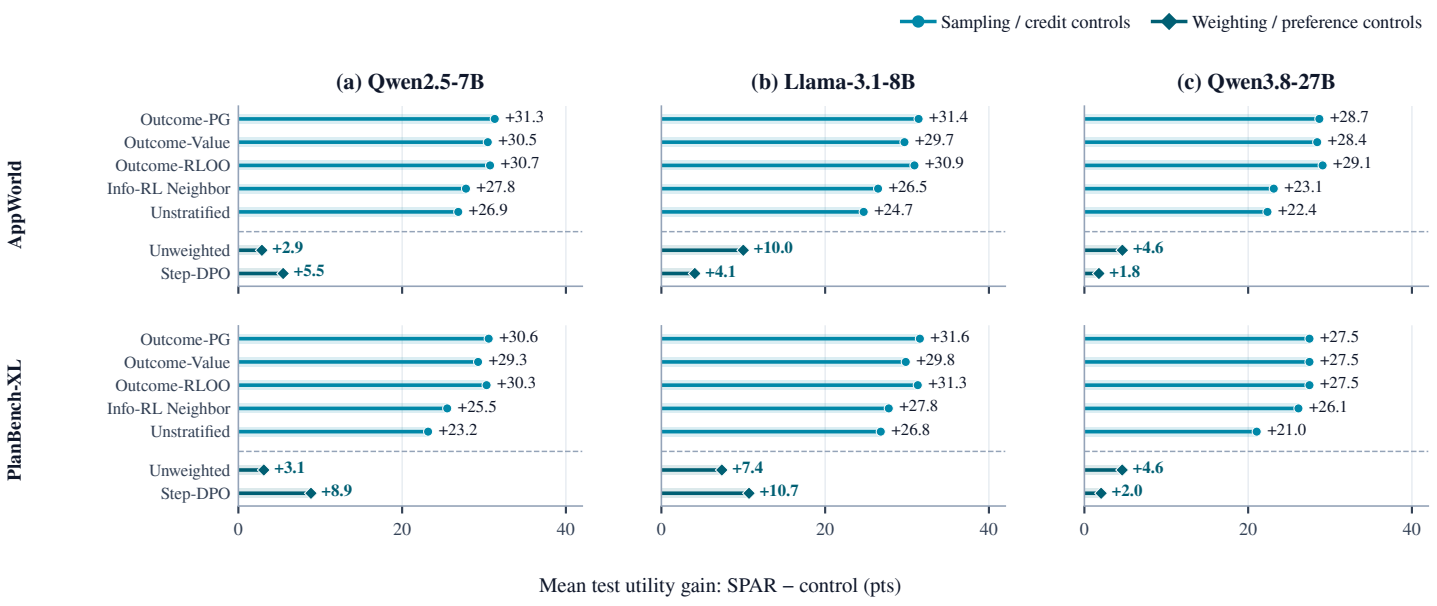
Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL
Frozen Base	34.92 ±1.37	35.93 ±3.90	36.90 ±1.57	36.67 ±4.44	46.03 ±1.37	45.56 ±5.56
AREW / T3	37.70 ±1.72	38.52 ±2.80	38.49 ±1.24	42.22 ±5.88	50.79 ±1.24	52.22 ±4.84
GiGPO	42.46 ±4.47	43.70 ±5.13	45.44 ±3.87	47.41 ±7.40	55.16 ±0.91	57.04 ±5.13
InfoPO	50.40 ±4.47	55.56 ±5.88	55.16 ±0.34	58.52 ±5.70	74.60 ±10.07	74.81 ±9.71
WorldEvolver	50.40 ±2.68	62.59 ±3.21	53.37 ±0.69	64.07 ±4.63	63.10 ±1.19	74.07 ±3.21
PoS / EnvProbe	72.42 ±0.34	70.00 ±2.94	80.56 ±0.91	81.85 ±2.57	**97.22** ±1.50	**97.04** ±0.64
Step-DPO	79.37 ±10.59	75.93 ±10.68	**83.53** ±9.42	79.63 ±9.71	96.63 ±5.84	94.81 ±5.25
Outcome-PG	41.07 ±2.15	44.07 ±4.49	43.06 ±2.25	48.89 ±9.62	56.94 ±4.96	57.41 ±4.49
Outcome-Value	42.26 ±3.90	45.93 ±6.12	45.63 ±3.28	51.48 ±8.04	57.34 ±4.51	57.41 ±4.49
Outcome-RLOO	41.87 ±3.38	44.44 ±3.85	43.85 ±3.96	49.26 ±8.98	56.35 ±5.72	57.41 ±4.49
Info-RL Neighbor	46.23 ±4.81	51.48 ±8.19	50.40 ±3.00	54.44 ±6.76	65.08 ±5.78	59.63 ±5.48
Unstratified	47.62 ±3.31	54.81 ±5.13	52.98 ±1.57	55.93 ±5.25	66.27 ±2.09	67.04 ±5.48
Unweighted	**88.49** ±19.93	**90.37** ±11.13	77.78 ±19.33	**90.00** ±15.40	96.43 ±6.19	94.81 ±7.06
SPAR	**88.49** ±17.91	**91.48** ±6.42	**90.28** ±16.84	**98.15** ±1.28	**99.60** ±0.69	**97.78** ±1.92

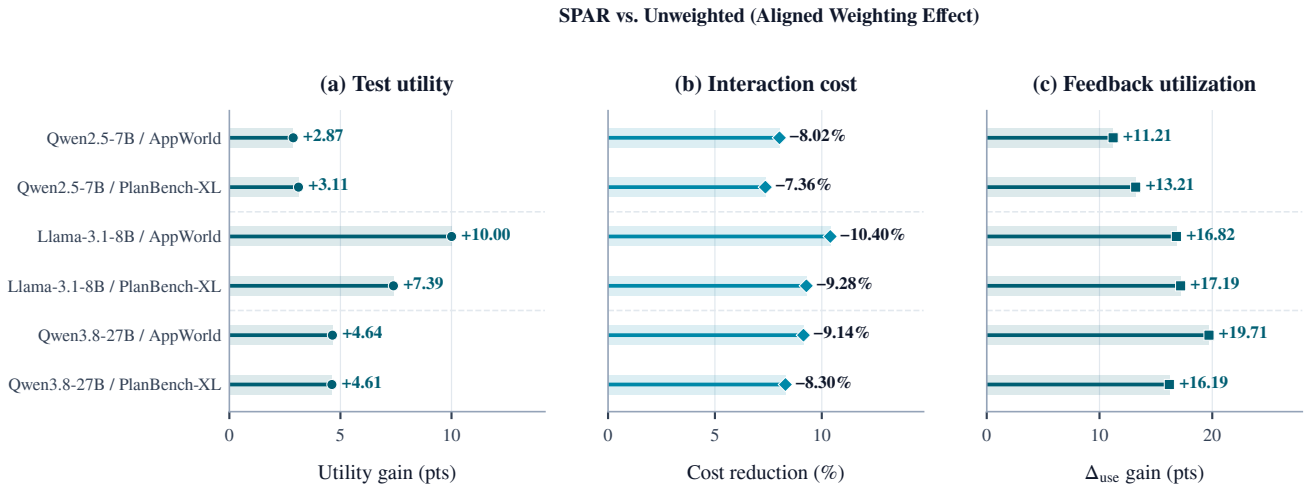

vs. Frozen Base
vs. Unweighted
vs. Best peer
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Frozen Base
Unweighted
Best peer
+35.47
+2.87
+5.47 (Step-DPO)
+35.59
+10.00
+4.08 (Step-DPO)
+36.06
+4.64
+1.77 (Step-DPO)
+36.10
+3.11
+8.87 (Step-DPO)
+39.95
+7.39
+10.50 (PoS / EnvProbe)
+35.58
+4.61
+2.04 (Step-DPO)
0
20
40
60
Mean test utility gain: SPAR − reference (pts)

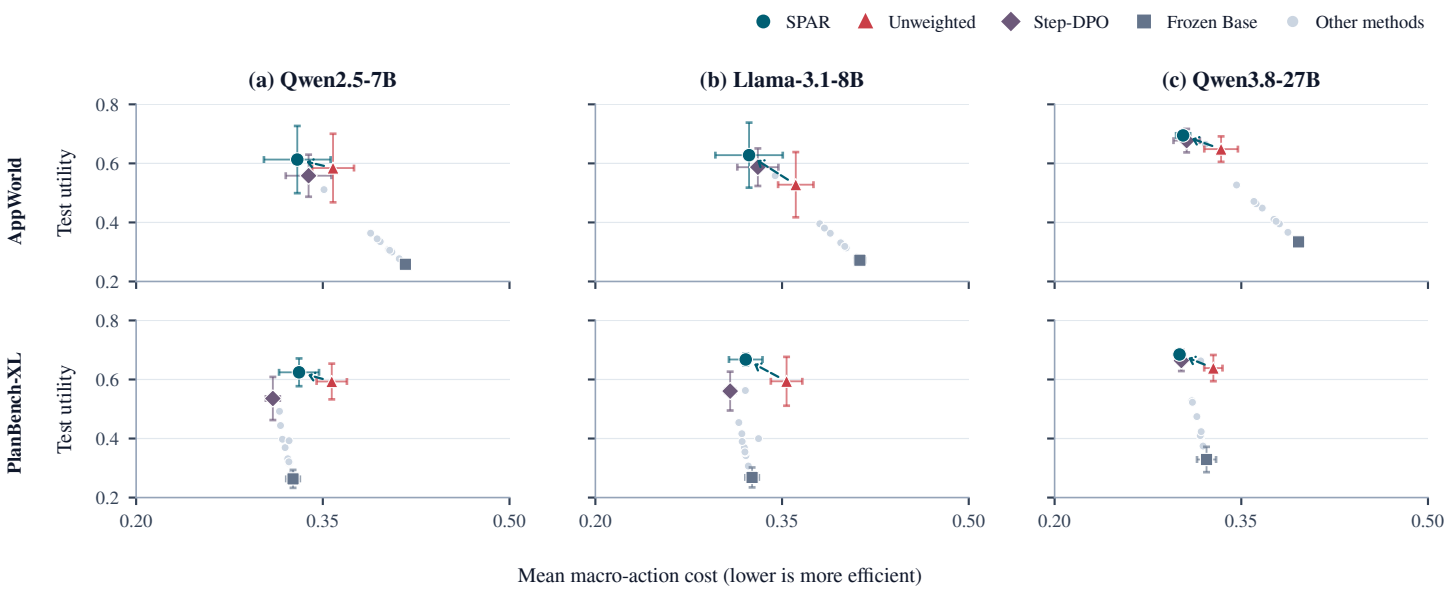

Qwen2.5-7B
Seed 42
Seed 43
Seed 44
Mean ± SD
SPAR mean
(a) AppWorld
(b) PlanBench-XL
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.613
0.624
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$
Test utility $\mathbb{E}[U_B]$

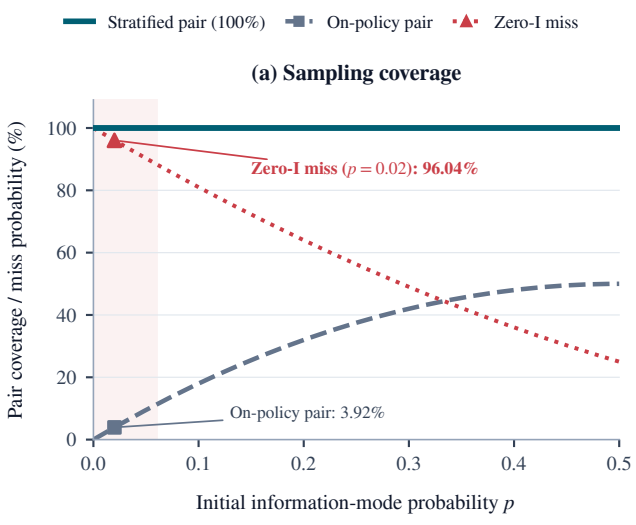

Llama-3.1-8B
Seed 42
Seed 43
Seed 44
Mean ± SD
SPAR mean
(a) AppWorld
(b) PlanBench-XL
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.628
0.668
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$
Test utility $\mathbb{E}[U_B]$


Qwen3.8-27B
Seed 42
Seed 43
Seed 44
Mean ± SD
SPAR mean
(a) AppWorld
(b) PlanBench-XL
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.695
0.685
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$
Test utility $\mathbb{E}[U_B]$


Sampling / credit controls
Weighting / preference controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
Step-DPO
+31.3
+30.5
+30.7
+27.8
+26.9
+2.9
+5.5
+31.4
+29.7
+30.9
+26.5
+24.7
+10.0
+4.1
+28.7
+28.4
+29.1
+23.1
+22.4
+4.6
+1.8
+30.6
+29.3
+30.3
+25.5
+23.2
+3.1
+8.9
+31.6
+29.8
+31.3
+27.8
+26.8
+7.4
+10.7
+27.5
+27.5
+27.5
+26.1
+21.0
+4.6
+2.0
0
20
40
Mean test utility gain: SPAR − control (pts)


SPAR vs. Unweighted (Aligned Weighting Effect)
(a) Test utility
(b) Interaction cost
(c) Feedback utilization
Qwen2.5-7B / AppWorld
Qwen2.5-7B / PlanBench-XL
Llama-3.1-8B / AppWorld
Llama-3.1-8B / PlanBench-XL
Qwen3.8-27B / AppWorld
Qwen3.8-27B / PlanBench-XL
+2.87
+3.11
+10.00
+7.39
+4.64
+4.61
−8.02%
−7.36%
−10.40%
−9.28%
−9.14%
−8.30%
+11.21
+13.21
+16.82
+17.19
+19.71
+16.19
0
5
10
0
5
10
0
10
20
Utility gain (pts)
Cost reduction (%)
Δuse gain (pts)


SPAR
Unweighted
Step-DPO
Frozen Base
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Test utility
0.8
0.6
0.4
0.2
0.20
0.35
0.50
Mean macro-action cost (lower is more efficient)

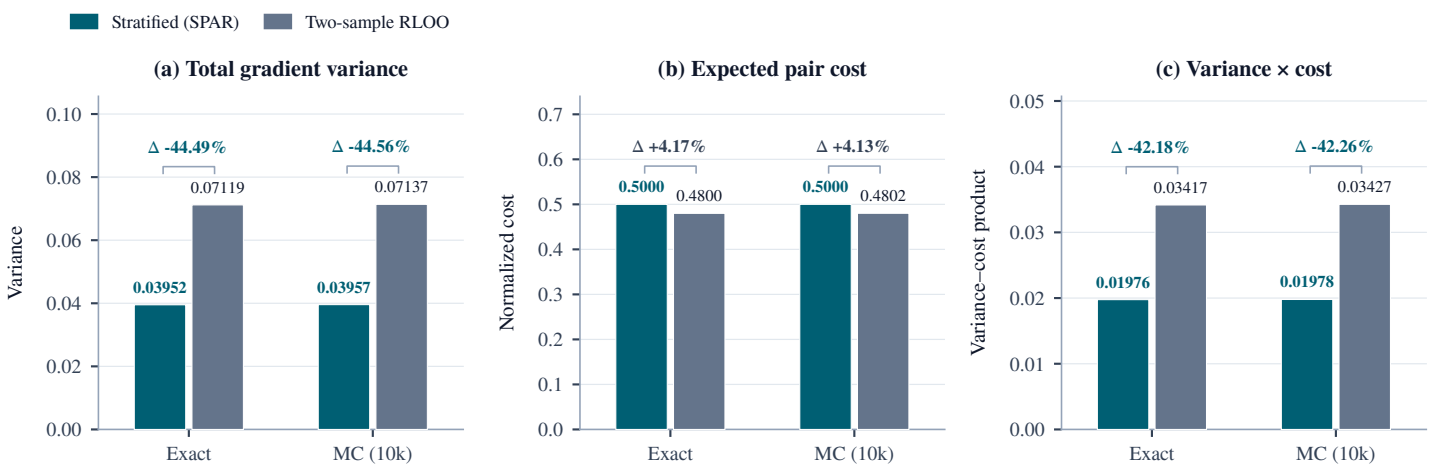


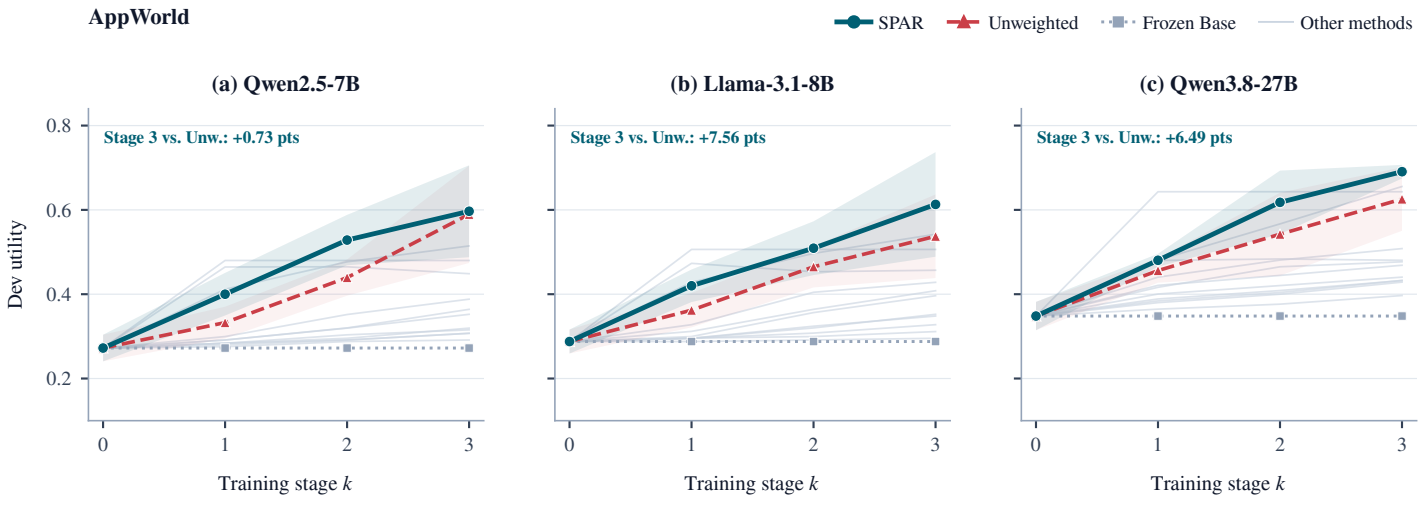
(b) Gradient agreement (MC: 10,000)

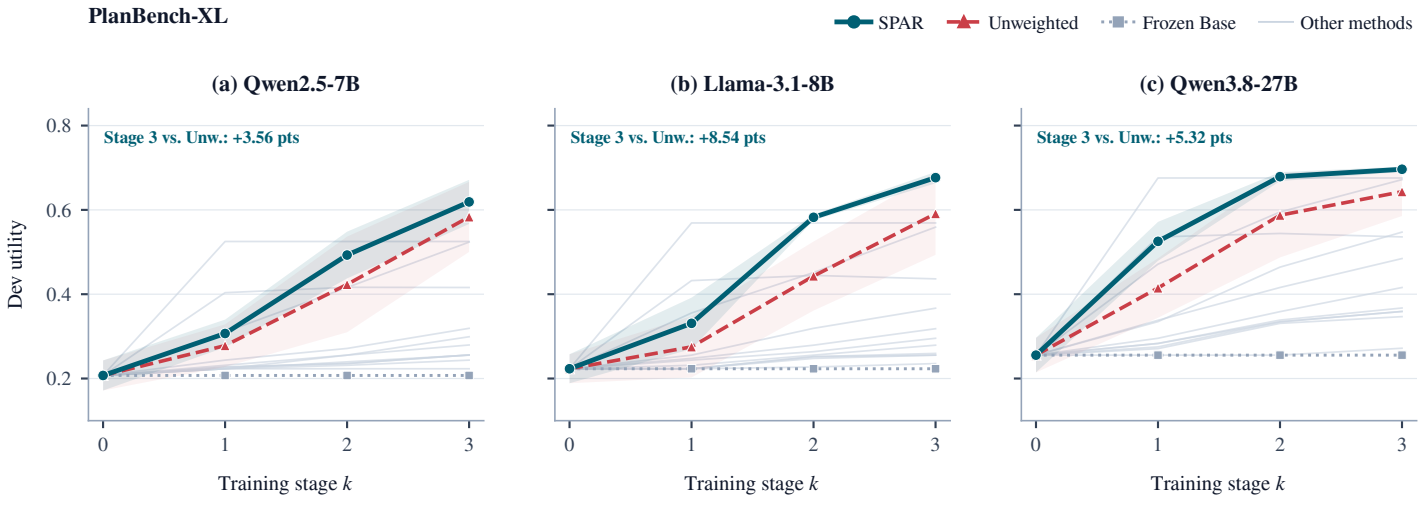
Objective $J(\theta_0)$ **58/125 = 0.4640**

Variance identity: LHS = RHS **0.0164736**

Estimator	g_1	g_2	g_3
Analytical	+0.03840	+0.04480	+0.00000
Stratified exact	**+0.03840**	**+0.04480**	**+0.00000**
RLOO exact	+0.03840	+0.04480	+0.00000
Autograd	+0.03840	+0.04480	+0.00000
Stratified MC	**+0.03814**	**+0.04439**	**-0.00031**
RLOO MC	+0.03736	+0.04529	-0.00164


Stratified (SPAR)
Two-sample RLOO
(a) Total gradient variance
Variance
0.10
0.08
0.06
0.04
0.02
0.00
Δ -44.49%
Δ -44.56%
0.07119
0.07137
0.03952
0.03957
Exact
MC (10k)
(b) Expected pair cost
Normalized cost
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Δ +4.17%
Δ +4.13%
0.5000
0.4800
0.5000
0.4802
Exact
MC (10k)
(c) Variance × cost
Variance–cost product
0.05
0.04
0.03
0.02
0.01
0.00
Δ -42.18%
Δ -42.26%
0.03417
0.03427
0.01976
0.01978
Exact
MC (10k)


AppWorld
SPAR
Unweighted
Frozen Base
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Stage 3 vs. Unw.: +0.73 pts
Stage 3 vs. Unw.: +7.56 pts
Stage 3 vs. Unw.: +6.49 pts
Dev utility
0.8
0.6
0.4
0.2
0
1
2
3
Training stage k


PlanBench-XL
SPAR
Unweighted
Frozen Base
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Stage 3 vs. Unw.: +3.56 pts
Stage 3 vs. Unw.: +8.54 pts
Stage 3 vs. Unw.: +5.32 pts
Dev utility
0.8
0.6
0.4
0.2
0
1
2
3
Training stage k

(a) AppWorld — Stages 0 and 1

Mean ± SD

Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	Stage 0	Stage 1	Stage 0	Stage 1	Stage 0	Stage 1
Frozen Base	0.2723 ±0.0316	0.2723 ±0.0316	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3482 ±0.0337
AREW / T3	0.2723 ±0.0316	0.2754 ±0.0282	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3646 ±0.0352
GiGPO	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.3887 ±0.0397
InfoPO	0.2723 ±0.0316	0.2995 ±0.0162	0.2877 ±0.0285	0.3277 ±0.0139	0.3482 ±0.0337	0.4405 ±0.0330
WorldEvolver	0.2723 ±0.0316	0.4646 ±0.0070	0.2877 ±0.0285	0.4732 ±0.0253	0.3482 ±0.0337	0.4800 ±0.0094
PoS / EnvProbe	0.2723 ±0.0316	**0.4800** ±0.0158	0.2877 ±0.0285	**0.5062** ±0.0289	0.3482 ±0.0337	**0.6430** ±0.0104
Step-DPO	0.2723 ±0.0316	0.4118 ±0.0455	0.2877 ±0.0285	0.4282 ±0.0440	0.3482 ±0.0337	0.4809 ±0.0151
Outcome-PG	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2959 ±0.0217	0.3482 ±0.0337	0.3841 ±0.0452
Outcome-Value	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.4005 ±0.0464
Outcome-RLOO	0.2723 ±0.0316	0.2795 ±0.0352	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.3800 ±0.0517
Info-RL Neighbor	0.2723 ±0.0316	0.2913 ±0.0233	0.2877 ±0.0285	0.2995 ±0.0162	0.3482 ±0.0337	0.4209 ±0.0330
Unstratified	0.2723 ±0.0316	0.2913 ±0.0233	0.2877 ±0.0285	0.3118 ±0.0277	0.3482 ±0.0337	0.4164 ±0.0327
Unweighted	0.2723 ±0.0316	0.3326 ±0.0377	0.2877 ±0.0285	0.3621 ±0.0408	0.3482 ±0.0337	0.4560 ±0.0287
SPAR	0.2723 ±0.0316	0.4000 ±0.0506	0.2877 ±0.0285	0.4200 ±0.0386	0.3482 ±0.0337	0.4805 ±0.0154

(b) AppWorld — Stages 2 and 3

Mean ± SD

Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	Stage 2	Stage 3	Stage 2	Stage 3	Stage 2	Stage 3
Frozen Base	0.2723 ±0.0316	0.2723 ±0.0316	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3482 ±0.0337
AREW / T3	0.2877 ±0.0285	0.2913 ±0.0267	0.2913 ±0.0233	0.2954 ±0.0209	0.3764 ±0.0472	0.3968 ±0.0285
GiGPO	0.3036 ±0.0344	0.3154 ±0.0558	0.3241 ±0.0602	0.3482 ±0.0725	0.4087 ±0.0397	0.4323 ±0.0275
InfoPO	0.3518 ±0.0867	0.3882 ±0.0686	0.4046 ±0.0416	0.4282 ±0.0330	0.4809 ±0.0151	0.5082 ±0.0646
WorldEvolver	0.4650 ±0.0199	0.4487 ±0.0344	0.4532 ±0.0145	0.4568 ±0.0395	0.4836 ±0.0144	0.4805 ±0.0155
PoS / EnvProbe	0.4800 ±0.0158	0.4800 ±0.0158	0.5062 ±0.0289	0.5062 ±0.0289	**0.6430** ±0.0104	0.6430 ±0.0104
Step-DPO	0.4768 ±0.0200	0.5145 ±0.0588	0.4968 ±0.0438	0.5422 ±0.0794	0.5668 ±0.0853	0.6554 ±0.0772
Outcome-PG	0.2913 ±0.0233	0.3073 ±0.0308	0.2995 ±0.0162	0.3113 ±0.0277	0.4041 ±0.0644	0.4327 ±0.0510
Outcome-Value	0.2954 ±0.0304	0.3195 ±0.0521	0.3200 ±0.0517	0.3523 ±0.0663	0.4205 ±0.0445	0.4405 ±0.0452
Outcome-RLOO	0.2913 ±0.0233	0.3073 ±0.0308	0.3077 ±0.0304	0.3277 ±0.0663	0.4000 ±0.0698	0.4287 ±0.0568
Info-RL Neighbor	0.3195 ±0.0521	0.3523 ±0.0671	0.3564 ±0.0704	0.3964 ±0.0518	0.4446 ±0.0395	0.4687 ±0.0324
Unstratified	0.3200 ±0.0533	0.3641 ±0.0583	0.3641 ±0.0710	0.4082 ±0.0389	0.4650 ±0.0208	0.4768 ±0.0080
Unweighted	0.4394 ±0.0430	0.5895 ±0.1155	0.4650 ±0.0490	0.5374 ±0.0991	0.5423 ±0.0984	0.6257 ±0.0754
SPAR	**0.5284** ±0.0598	**0.5967** ±0.1090	**0.5091** ±0.0638	**0.6130** ±0.1241	0.6179 ±0.0753	**0.6906** ±0.0162

(a) PlanBench-XL — Stages 0 and 1

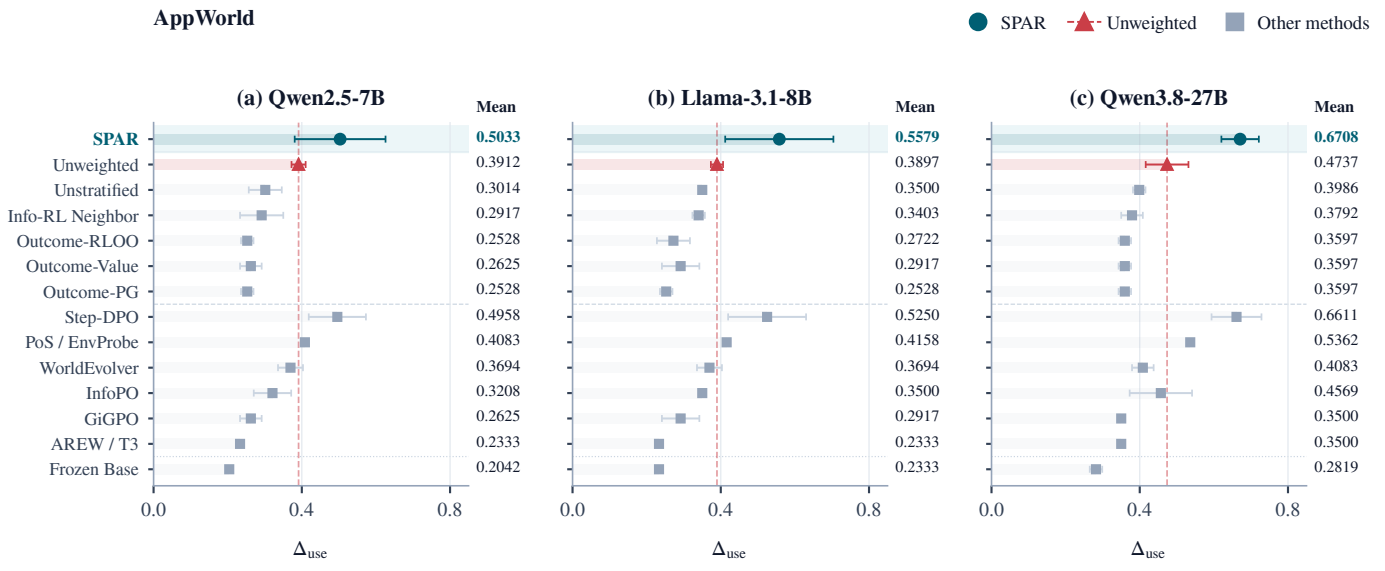
Mean ± SD

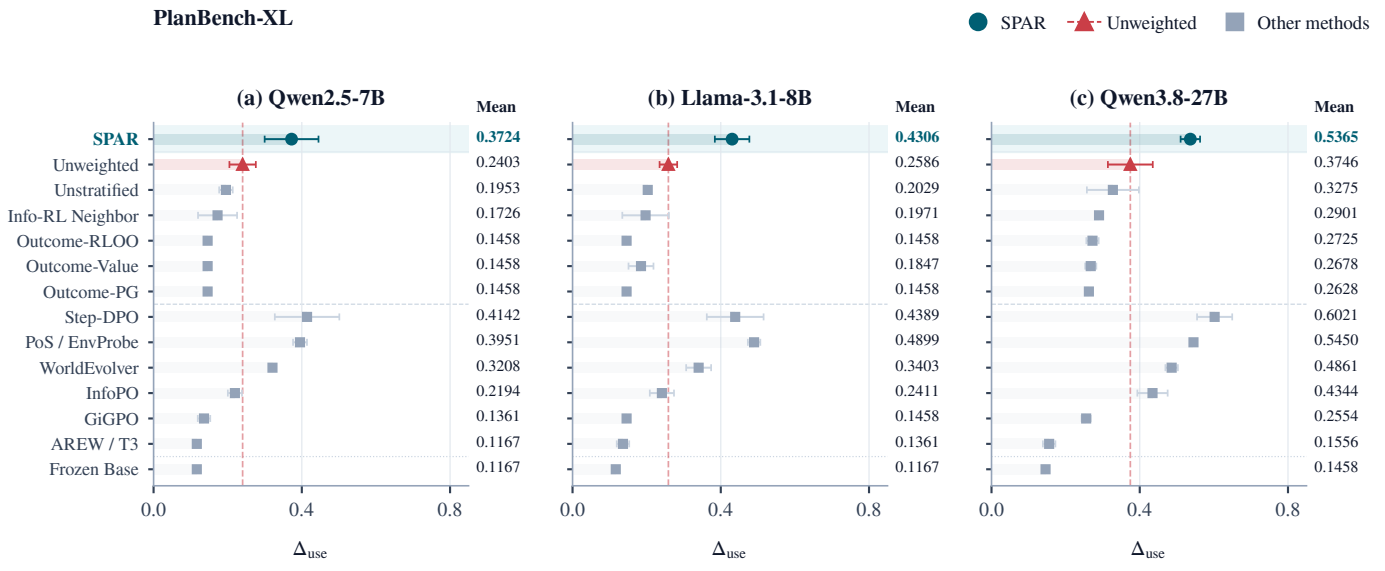
Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	Stage 0	Stage 1	Stage 0	Stage 1	Stage 0	Stage 1
Frozen Base	0.2073 ±0.0357	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411
AREW / T3	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411
GiGPO	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2709 ±0.0550
InfoPO	0.2073 ±0.0357	0.2436 ±0.0359	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411	0.3350 ±0.0281
WorldEvolver	0.2073 ±0.0357	0.4036 ±0.0217	0.2232 ±0.0344	0.4323 ±0.0206	0.2554 ±0.0411	0.5359 ±0.0199
PoS / EnvProbe	0.2073 ±0.0357	**0.5250** ±0.0111	0.2232 ±0.0344	**0.5690** ±0.0087	0.2554 ±0.0411	**0.6757** ±0.0063
Step-DPO	0.2073 ±0.0357	0.3104 ±0.0605	0.2232 ±0.0344	0.3554 ±0.0743	0.2554 ±0.0411	0.4713 ±0.0744
Outcome-PG	0.2073 ±0.0357	0.2273 ±0.0410	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2750 ±0.0618
Outcome-Value	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2313 ±0.0281	0.2554 ±0.0411	0.2832 ±0.0529
Outcome-RLOO	0.2073 ±0.0357	0.2273 ±0.0410	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2832 ±0.0529
Info-RL Neighbor	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2436 ±0.0359	0.2554 ±0.0411	0.2954 ±0.0432
Unstratified	0.2073 ±0.0357	0.2313 ±0.0404	0.2232 ±0.0344	0.2477 ±0.0531	0.2554 ±0.0411	0.3381 ±0.0666
Unweighted	0.2073 ±0.0357	0.2780 ±0.0504	0.2232 ±0.0344	0.2751 ±0.0726	0.2554 ±0.0411	0.4147 ±0.0706
SPAR	0.2073 ±0.0357	0.3068 ±0.0330	0.2232 ±0.0344	0.3309 ±0.0610	0.2554 ±0.0411	0.5254 ±0.0462

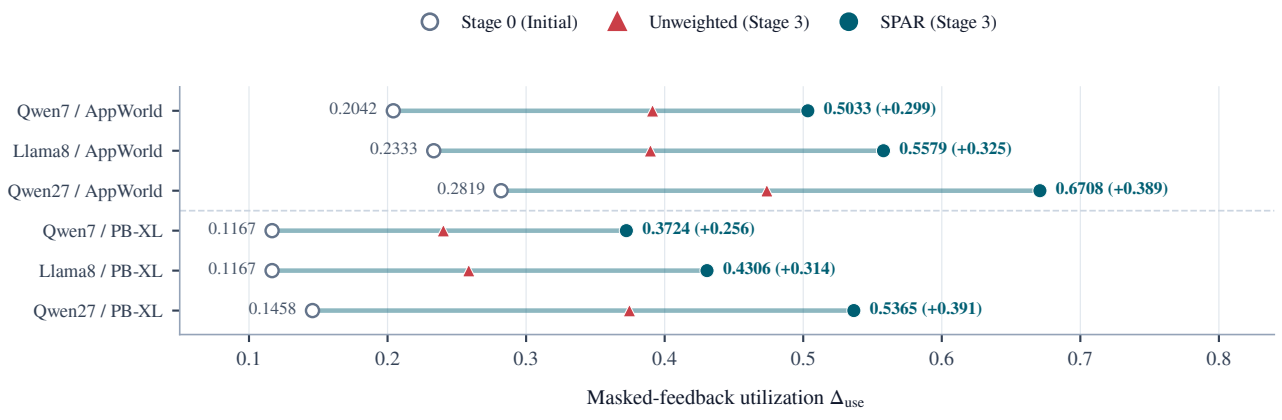
(b) PlanBench-XL — Stages 2 and 3

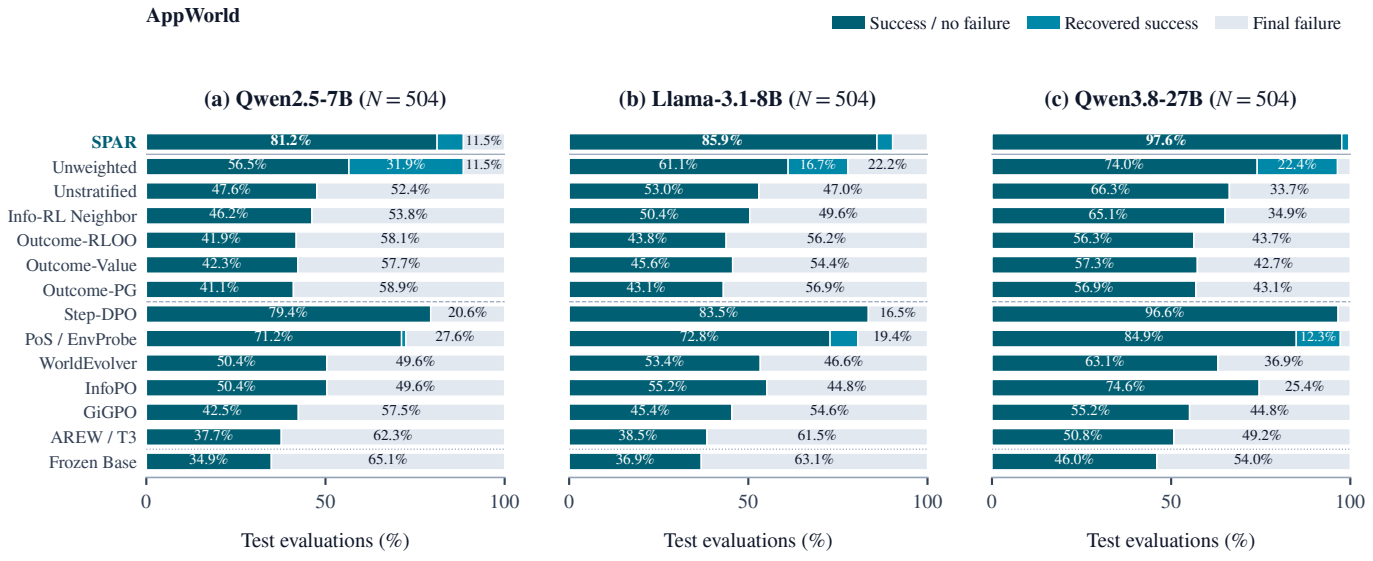
Mean ± SD

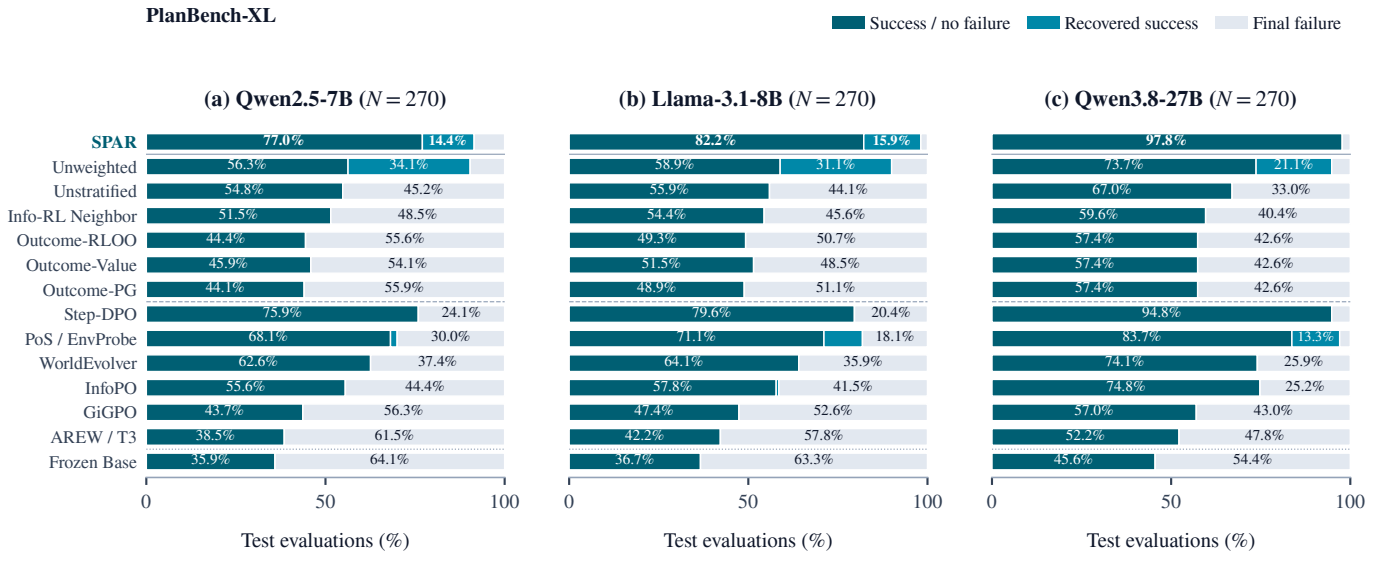
Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	Stage 2	Stage 3	Stage 2	Stage 3	Stage 2	Stage 3
Frozen Base	0.2073 ±0.0357	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411
AREW / T3	0.2232 ±0.0344	0.2232 ±0.0344	0.2273 ±0.0344	0.2354 ±0.0223	0.2554 ±0.0411	0.2718 ±0.0392
GiGPO	0.2313 ±0.0404	0.2436 ±0.0476	0.2477 ±0.0289	0.2554 ±0.0411	0.3309 ±0.0383	0.3468 ±0.0215
InfoPO	0.2713 ±0.0494	0.3191 ±0.0494	0.3191 ±0.0494	0.3668 ±0.0483	0.4642 ±0.0709	0.5473 ±0.0284
WorldEvolver	0.4164 ±0.0168	0.4159 ±0.0312	0.4446 ±0.0123	0.4364 ±0.0267	0.5441 ±0.0131	0.5359 ±0.0199
PoS / EnvProbe	**0.5250** ±0.0111	0.5250 ±0.0111	0.5690 ±0.0087	0.5690 ±0.0087	0.6757 ±0.0063	0.6757 ±0.0063
Step-DPO	0.4159 ±0.0424	0.5236 ±0.0884	0.4513 ±0.0517	0.5591 ±0.0660	0.5950 ±0.0142	0.6718 ±0.0488
Outcome-PG	0.2354 ±0.0467	0.2559 ±0.0404	0.2518 ±0.0344	0.2554 ±0.0365	0.3345 ±0.0421	0.3591 ±0.0403
Outcome-Value	0.2395 ±0.0466	0.2554 ±0.0411	0.2554 ±0.0411	0.2791 ±0.0255	0.3386 ±0.0470	0.3673 ±0.0524
Outcome-RLOO	0.2354 ±0.0467	0.2559 ±0.0404	0.2518 ±0.0344	0.2595 ±0.0392	0.3345 ±0.0421	0.3591 ±0.0459
Info-RL Neighbor	0.2554 ±0.0411	0.2795 ±0.0468	0.2636 ±0.0434	0.2954 ±0.0328	0.3586 ±0.0412	0.4159 ±0.0163
Unstratified	0.2554 ±0.0411	0.2991 ±0.0351	0.2791 ±0.0566	0.3181 ±0.0487	0.4159 ±0.0831	0.4846 ±0.0688
Unweighted	0.4232 ±0.1135	0.5834 ±0.0831	0.4430 ±0.0820	0.5911 ±0.0978	0.5874 ±0.1002	0.6432 ±0.0577
SPAR	0.4929 ±0.0555	**0.6189** ±0.0524	**0.5823** ±0.0071	**0.6765** ±0.0136	**0.6788** ±0.0107	**0.6964** ±0.0063


AppWorld
SPAR
Unweighted
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Mean
SPAR 0.5033 0.5579 0.6708
Unweighted 0.3912 0.3897 0.4737
Unstratified 0.3014 0.3500 0.3986
Info-RL Neighbor 0.2917 0.3403 0.3792
Outcome-RLOO 0.2528 0.2722 0.3597
Outcome-Value 0.2625 0.2917 0.3597
Outcome-PG 0.2528 0.2528 0.3597
Step-DPO 0.4958 0.5250 0.6611
PoS / EnvProbe 0.4083 0.4158 0.5362
WorldEvolver 0.3694 0.3694 0.4083
InfoPO 0.3208 0.3500 0.4569
GiGPO 0.2625 0.2917 0.3500
AREW / T3 0.2333 0.2333 0.3500
Frozen Base 0.2042 0.2333 0.2819
0.0
0.4
0.8
Δuse


PlanBench-XL
SPAR
Unweighted
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Mean
SPAR 0.3724 0.4306 0.5365
Unweighted 0.2403 0.2586 0.3746
Unstratified 0.1953 0.2029 0.3275
Info-RL Neighbor 0.1726 0.1971 0.2901
Outcome-RLOO 0.1458 0.1458 0.2725
Outcome-Value 0.1458 0.1847 0.2678
Outcome-PG 0.1458 0.1458 0.2628
Step-DPO 0.4142 0.4389 0.6021
PoS / EnvProbe 0.3951 0.4899 0.5450
WorldEvolver 0.3208 0.3403 0.4861
InfoPO 0.2194 0.2411 0.4344
GiGPO 0.1361 0.1458 0.2554
AREW / T3 0.1167 0.1361 0.1556
Frozen Base 0.1167 0.1167 0.1458
0.0
0.4
0.8
Δuse


Stage 0 (Initial)
Unweighted (Stage 3)
SPAR (Stage 3)
Qwen7 / AppWorld
Llama8 / AppWorld
Qwen27 / AppWorld
Qwen7 / PB-XL
Llama8 / PB-XL
Qwen27 / PB-XL
0.2042
0.5033 (+0.299)
0.2333
0.5579 (+0.325)
0.2819
0.6708 (+0.389)
0.1167
0.3724 (+0.256)
0.1167
0.4306 (+0.314)
0.1458
0.5365 (+0.391)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Masked-feedback utilization Δ_{use}


AppWorld
Success / no failure
Recovered success
Final failure
(a) Qwen2.5-7B (N = 504)
(b) Llama-3.1-8B (N = 504)
(c) Qwen3.8-27B (N = 504)
SPAR
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
81.2%
11.5%
56.5%
31.9%
11.5%
47.6%
52.4%
46.2%
53.8%
41.9%
58.1%
42.3%
57.7%
41.1%
58.9%
79.4%
20.6%
71.2%
27.6%
50.4%
49.6%
50.4%
49.6%
42.5%
57.5%
37.7%
62.3%
34.9%
65.1%
85.9%
61.1%
16.7%
22.2%
53.0%
47.0%
50.4%
49.6%
43.8%
56.2%
45.6%
54.4%
43.1%
56.9%
83.5%
16.5%
72.8%
19.4%
53.4%
46.6%
55.2%
44.8%
45.4%
54.6%
38.5%
61.5%
36.9%
63.1%
97.6%
74.0%
22.4%
66.3%
33.7%
65.1%
34.9%
56.3%
43.7%
57.3%
42.7%
56.9%
43.1%
96.6%
84.9%
12.3%
63.1%
36.9%
74.6%
25.4%
55.2%
44.8%
50.8%
49.2%
46.0%
54.0%
0
50
100
Test evaluations (%)


PlanBench-XL
Success / no failure
Recovered success
Final failure
(a) Qwen2.5-7B (N = 270)
(b) Llama-3.1-8B (N = 270)
(c) Qwen3.8-27B (N = 270)
SPAR
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
77.0%
14.4%
56.3%
34.1%
54.8%
45.2%
51.5%
48.5%
44.4%
55.6%
45.9%
54.1%
44.1%
55.9%
75.9%
24.1%
68.1%
30.0%
62.6%
37.4%
55.6%
44.4%
43.7%
56.3%
38.5%
61.5%
35.9%
64.1%
82.2%
15.9%
58.9%
31.1%
55.9%
44.1%
54.4%
45.6%
49.3%
50.7%
51.5%
48.5%
48.9%
51.1%
79.6%
20.4%
71.1%
18.1%
64.1%
35.9%
57.8%
41.5%
47.4%
52.6%
42.2%
57.8%
36.7%
63.3%
97.8%
73.7%
21.1%
67.0%
33.0%
59.6%
40.4%
57.4%
42.6%
57.4%
42.6%
57.4%
42.6%
94.8%
83.7%
13.3%
74.1%
25.9%
74.8%
25.2%
57.0%
43.0%
52.2%
47.8%
45.6%
54.4%
0
50
100
Test evaluations (%)

(a) Masked-feedback utilization Δ_{use} (fixed 24-task diagnostic)

Δ_{use} mean ± SD

Method	AppWorld			PlanBench-XL		
	Qwen7	Llama8	Qwen27	Qwen7	Llama8	Qwen27
Frozen Base	0.2042 ±0.0000	0.2333 ±0.0000	0.2819 ±0.0168	0.1167 ±0.0000	0.1167 ±0.0000	0.1458 ±0.0000
AREW / T3	0.2333 ±0.0000	0.2333 ±0.0000	0.3500 ±0.0000	0.1167 ±0.0000	0.1361 ±0.0168	0.1556 ±0.0168
GiGPO	0.2625 ±0.0292	0.2917 ±0.0505	0.3500 ±0.0000	0.1361 ±0.0168	0.1458 ±0.0000	0.2554 ±0.0123
InfoPO	0.3208 ±0.0505	0.3500 ±0.0000	0.4569 ±0.0842	0.2194 ±0.0191	0.2411 ±0.0327	0.4344 ±0.0409
WorldEvolver	0.3694 ±0.0337	0.3694 ±0.0337	0.4083 ±0.0292	0.3208 ±0.0000	0.3403 ±0.0337	0.4861 ±0.0168
PoS / EnvProbe	0.4083 ±0.0000	0.4158 ±0.0038	0.5362 ±0.0000	0.3951 ±0.0188	**0.4899** ±0.0168	0.5450 ±0.0022
Step-DPO	0.4958 ±0.0772	0.5250 ±0.1052	0.6611 ±0.0674	**0.4142** ±0.0871	0.4389 ±0.0766	**0.6021** ±0.0474
Outcome-PG	0.2528 ±0.0168	0.2528 ±0.0168	0.3597 ±0.0168	0.1458 ±0.0000	0.1458 ±0.0000	0.2628 ±0.0075
Outcome-Value	0.2625 ±0.0292	0.2917 ±0.0505	0.3597 ±0.0168	0.1458 ±0.0000	0.1847 ±0.0337	0.2678 ±0.0151
Outcome-RLOO	0.2528 ±0.0168	0.2722 ±0.0446	0.3597 ±0.0168	0.1458 ±0.0000	0.1458 ±0.0000	0.2725 ±0.0166
Info-RL Neighbor	0.2917 ±0.0583	0.3403 ±0.0168	0.3792 ±0.0292	0.1726 ±0.0527	0.1971 ±0.0628	0.2901 ±0.0041
Unstratified	0.3014 ±0.0446	0.3500 ±0.0000	0.3986 ±0.0168	0.1953 ±0.0183	0.2029 ±0.0036	0.3275 ±0.0702
Unweighted	0.3912 ±0.0191	0.3897 ±0.0165	0.4737 ±0.0577	0.2403 ±0.0356	0.2586 ±0.0237	0.3746 ±0.0606
SPAR	**0.5033** ±0.1226	**0.5579** ±0.1460	**0.6708** ±0.0505	0.3724 ±0.0726	0.4306 ±0.0466	0.5365 ±0.0262

(b) Conditional recovery counts and pooled recovery rates

Recovered / failure (%)

Method	AppWorld			PlanBench-XL		
	Qwen7	Llama8	Qwen27	Qwen7	Llama8	Qwen27
Frozen Base	0/328 (0.0%)	0/318 (0.0%)	0/272 (0.0%)	0/173 (0.0%)	0/171 (0.0%)	0/147 (0.0%)
AREW / T3	0/314 (0.0%)	0/310 (0.0%)	0/248 (0.0%)	0/166 (0.0%)	0/156 (0.0%)	0/129 (0.0%)
GiGPO	0/290 (0.0%)	0/275 (0.0%)	0/226 (0.0%)	0/152 (0.0%)	0/142 (0.0%)	0/116 (0.0%)
InfoPO	0/250 (0.0%)	0/226 (0.0%)	0/128 (0.0%)	0/120 (0.0%)	2/114 (1.8%)	0/68 (0.0%)
WorldEvolver	0/250 (0.0%)	0/235 (0.0%)	0/186 (0.0%)	0/101 (0.0%)	0/97 (0.0%)	0/70 (0.0%)
PoS / EnvProbe	6/145 (4.1%)	39/137 (28.5%)	62/76 (81.6%)	5/86 (5.8%)	29/78 (37.2%)	**36/44 (81.8%)**
Step-DPO	0/104 (0.0%)	0/83 (0.0%)	0/17 (0.0%)	0/65 (0.0%)	0/55 (0.0%)	0/14 (0.0%)
Outcome-PG	0/297 (0.0%)	0/287 (0.0%)	0/217 (0.0%)	0/151 (0.0%)	0/138 (0.0%)	0/115 (0.0%)
Outcome-Value	0/291 (0.0%)	0/274 (0.0%)	0/215 (0.0%)	0/146 (0.0%)	0/131 (0.0%)	0/115 (0.0%)
Outcome-RLOO	0/293 (0.0%)	0/283 (0.0%)	0/220 (0.0%)	0/150 (0.0%)	0/137 (0.0%)	0/115 (0.0%)
Info-RL Neighbor	0/271 (0.0%)	0/250 (0.0%)	0/176 (0.0%)	0/131 (0.0%)	0/123 (0.0%)	0/109 (0.0%)
Unstratified	0/264 (0.0%)	0/237 (0.0%)	0/170 (0.0%)	0/122 (0.0%)	0/119 (0.0%)	0/89 (0.0%)
Unweighted	**161/219 (73.5%)**	**84/196 (42.9%)**	**113/131 (86.3%)**	**92/118 (78.0%)**	84/111 (75.7%)	57/71 (80.3%)
SPAR	37/95 (38.9%)	22/71 (31.0%)	10/12 (83.3%)	39/62 (62.9%)	**43/48 (89.6%)**	0/6 (0.0%)

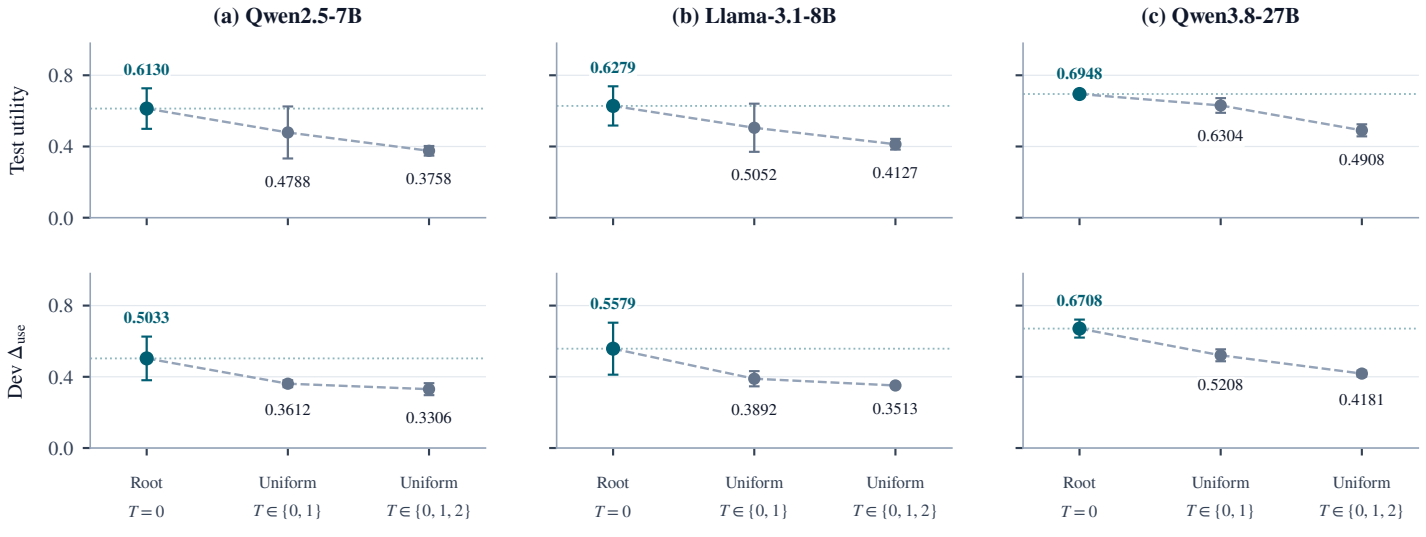

AppWorld
SPAR
Unweighted
Frozen Base
Default budget
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Tight budget: +2.53 pts
Tight budget: +6.12 pts
Tight budget: +6.92 pts
Dev utility
Dev success (%)
0.8
0.6
0.4
0.2
100
75
50
25
0.6×
1.0×
1.4×
Budget multiplier B/B_0

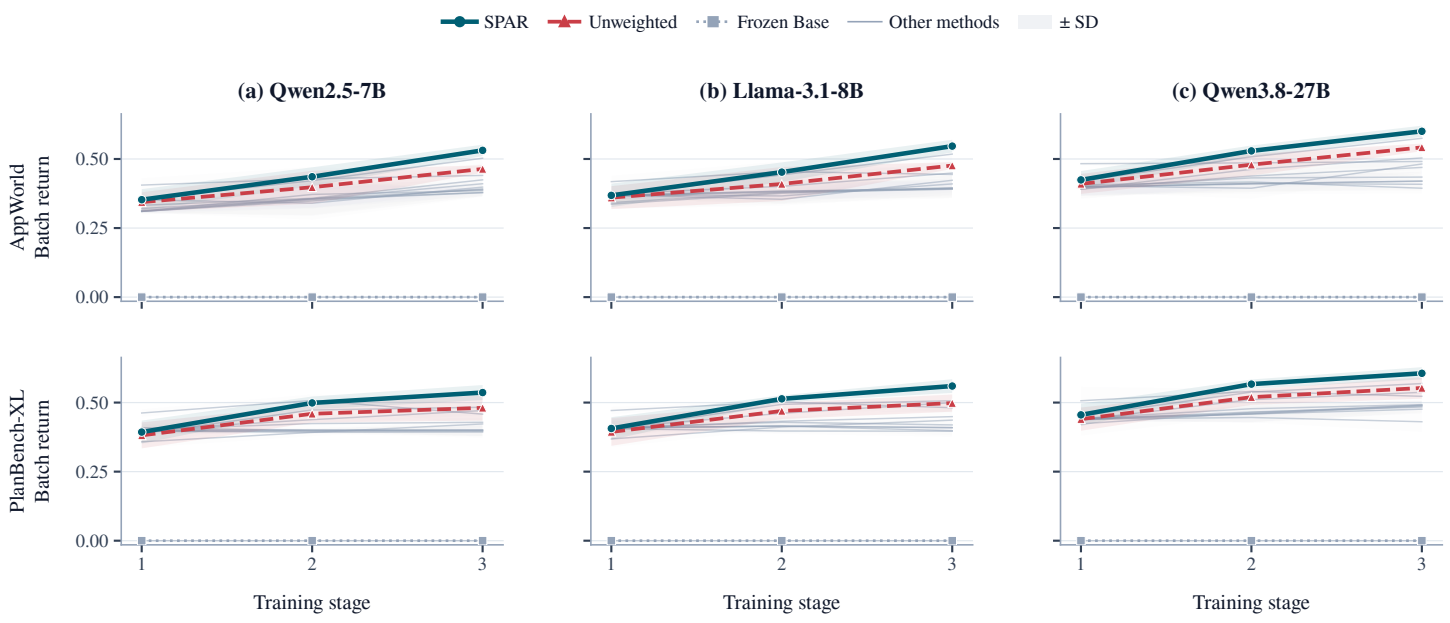
(a) AppWorld development utility across budgets

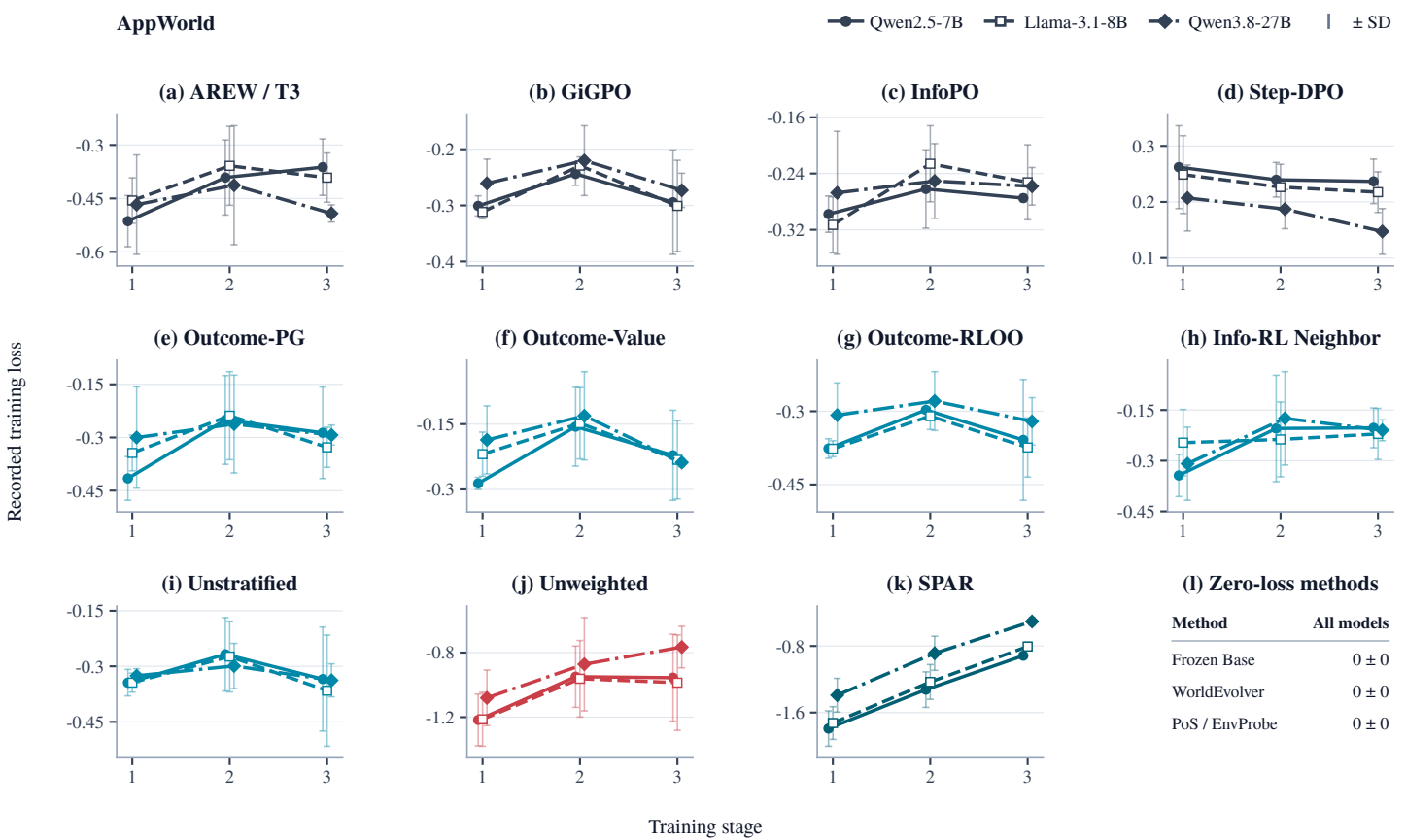
Dev utility	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
Method	**0.6×**	**1.0×**	**1.4×**	**0.6×**	**1.0×**	**1.4×**	**0.6×**	**1.0×**	**1.4×**
Frozen Base	0.2043	0.2723	0.3014	0.2144	0.2877	0.3191	0.2567	0.3482	0.3874
AREW / T3	0.2166	0.2913	0.3234	0.2195	0.2954	0.3280	0.2910	0.3968	0.4422
GiGPO	0.2333	0.3154	0.3506	0.2567	0.3482	0.3874	0.3150	0.4323	0.4825
InfoPO	0.2844	0.3882	0.4327	0.3121	0.4282	0.4779	0.3674	0.5082	0.5685
WorldEvolver	0.3267	0.4487	0.5009	0.3326	0.4568	0.5101	0.3485	0.4805	0.5370
PoS / EnvProbe	0.3478	0.4800	0.5367	0.3602	0.5062	0.5688	0.4363	0.6430	0.7317
Step-DPO	0.3702	0.5145	0.5764	0.3891	0.5422	0.6079	0.4686	0.6554	0.7355
Outcome-PG	0.2275	0.3073	0.3414	0.2304	0.3113	0.3460	0.3158	0.4327	0.4829
Outcome-Value	0.2363	0.3195	0.3552	0.2596	0.3523	0.3920	0.3209	0.4405	0.4917
Outcome-RLOO	0.2275	0.3073	0.3414	0.2421	0.3277	0.3644	0.3129	0.4287	0.4783
Info-RL Neighbor	0.2596	0.3523	0.3920	0.2903	0.3964	0.4419	0.3405	0.4687	0.5236
Unstratified	0.2676	0.3641	0.4054	0.2982	0.4082	0.4553	0.3464	0.4768	0.5327
Unweighted	0.3883	0.5895	0.6750	0.3678	0.5374	0.6094	0.4199	0.6257	0.7166
SPAR	**0.4136**	**0.5967**	**0.6752**	**0.4290**	**0.6130**	**0.6918**	**0.4891**	**0.6906**	**0.7837**

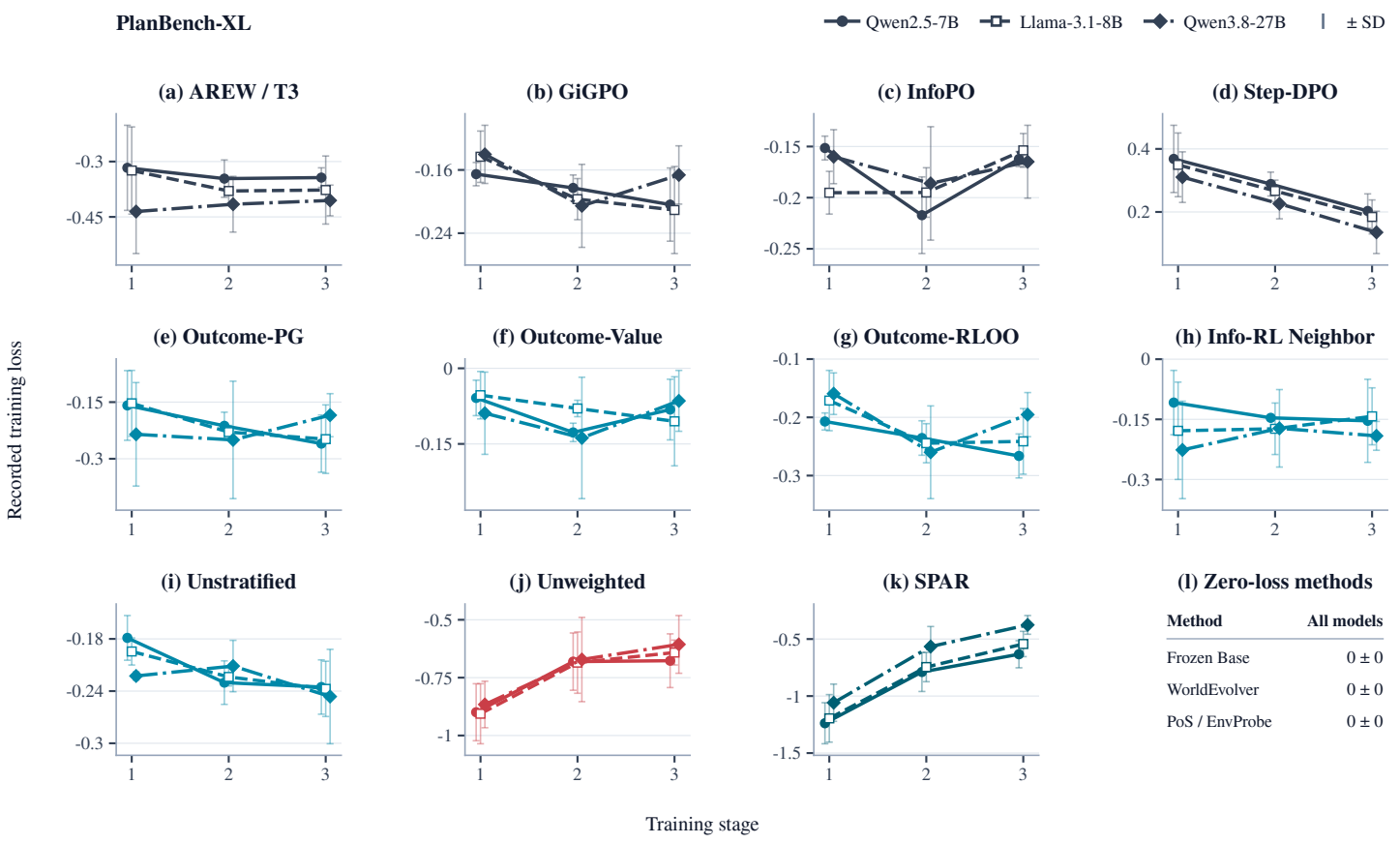
(b) AppWorld development success rate (%) across budgets

Dev success (%)	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
Method	**0.6×**	**1.0×**	**1.4×**	**0.6×**	**1.0×**	**1.4×**	**0.6×**	**1.0×**	**1.4×**
Frozen Base	37.43	37.43	37.43	39.77	39.77	39.77	48.54	48.54	48.54
AREW / T3	40.35	40.35	40.35	40.94	40.94	40.94	55.56	55.56	55.56
GiGPO	43.86	43.86	43.86	48.54	48.54	48.54	60.82	60.82	60.82
InfoPO	54.39	54.39	54.39	60.23	60.23	60.23	71.93	71.93	71.93
WorldEvolver	63.16	63.16	63.16	64.33	64.33	64.33	67.84	67.84	67.84
PoS / EnvProbe	67.84	67.84	67.84	72.51	72.51	72.51	95.32	95.32	95.32
Step-DPO	73.10	73.10	73.10	77.19	77.19	77.19	93.57	93.57	93.57
Outcome-PG	42.69	42.69	42.69	43.27	43.27	43.27	60.82	60.82	60.82
Outcome-Value	44.44	44.44	44.44	49.12	49.12	49.12	61.99	61.99	61.99
Outcome-RLOO	42.69	42.69	42.69	45.61	45.61	45.61	60.23	60.23	60.23
Info-RL Neighbor	49.12	49.12	49.12	55.56	55.56	55.56	66.08	66.08	66.08
Unstratified	50.88	50.88	50.88	57.31	57.31	57.31	67.25	67.25	67.25
Unweighted	**88.30**	**88.89**	**88.89**	78.36	78.95	78.95	91.81	92.98	93.57
SPAR	87.13	87.13	87.13	**88.89**	**88.89**	**88.89**	**97.66**	**98.83**	**100.00**


(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Test utility
Dev Δ_{use}
0.8
0.4
0.0
0.6130
0.4788
0.3758
0.6279
0.5052
0.4127
0.6948
0.6304
0.4908
0.5033
0.3612
0.3306
0.5579
0.3892
0.3513
0.6708
0.5208
0.4181
Root
$T = 0$
Uniform
$T \in \{0, 1\}$
Uniform
$T \in \{0, 1, 2\}$


SPAR
Unweighted
Frozen Base
Other methods
± SD
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
Batch return
PlanBench-XL
Batch return
0.00
0.25
0.50
1
2
3
Training stage


AppWorld
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
± SD
(a) AREW / T3
(b) GiGPO
(c) InfoPO
(d) Step-DPO
(e) Outcome-PG
(f) Outcome-Value
(g) Outcome-RLOO
(h) Info-RL Neighbor
(i) Unstratified
(j) Unweighted
(k) SPAR
(l) Zero-loss methods
Method
All models
Frozen Base
0 ± 0
WorldEvolver
0 ± 0
PoS / EnvProbe
0 ± 0
Recorded training loss
Training stage


PlanBench-XL
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
± SD
(a) AREW / T3
(b) GiGPO
(c) InfoPO
(d) Step-DPO
(e) Outcome-PG
(f) Outcome-Value
(g) Outcome-RLOO
(h) Info-RL Neighbor
(i) Unstratified
(j) Unweighted
(k) SPAR
(l) Zero-loss methods
Method
All models
Frozen Base
0 ± 0
WorldEvolver
0 ± 0
PoS / EnvProbe
0 ± 0
Recorded training loss
Training stage

(a) AppWorld — Training interaction work and total wall-clock time per seed

Method	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
	Env. steps	Tokens (k)	Wall (s)	Env. steps	Tokens (k)	Wall (s)	Env. steps	Tokens (k)	Wall (s)
Frozen Base	0	0	0	0	0	<0.01	0	0	0
AREW / T3	1434.0	489.0	30.65	1407.7	471.2	30.52	1405.7	479.7	7.88
GiGPO	1428.7	487.1	31.51	1401.3	468.2	68.98	1397.7	475.3	7.92
InfoPO	1426.0	488.0	33.26	1408.0	471.4	60.01	1371.0	460.9	8.95
WorldEvolver	1319.7	582.9	3.93	1309.3	573.1	3.98	1306.3	579.8	3.63
PoS / EnvProbe	0	0	0.59	0	0	0.65	0	0	0.60
Step-DPO	1362.0	448.4	40.13	1365.0	445.1	26.03	1366.3	453.9	10.12
Outcome-PG	1438.0	491.0	26.32	1402.7	469.2	38.96	1407.7	480.9	8.79
Outcome-Value	1434.3	490.1	35.87	1396.7	466.0	73.27	1387.0	469.9	7.38
Outcome-RLOO	1433.3	488.6	35.42	1397.3	465.0	39.54	1406.7	480.6	7.48
Info-RL Neighbor	1436.3	493.5	30.50	1417.0	476.1	45.34	1392.3	471.8	7.30
Unstratified	1435.7	489.9	15.04	1422.3	476.8	29.46	1381.0	462.1	6.45
Unweighted	1372.0	454.0	6.50	1369.3	446.4	6.85	1368.7	454.5	6.54
SPAR	**1367.3**	**450.5**	**28.81**	**1360.7**	**441.5**	**59.16**	**1366.7**	**453.6**	**8.20**

(b) AppWorld — Model encoding, gradient-update, and snapshot-restore time per seed

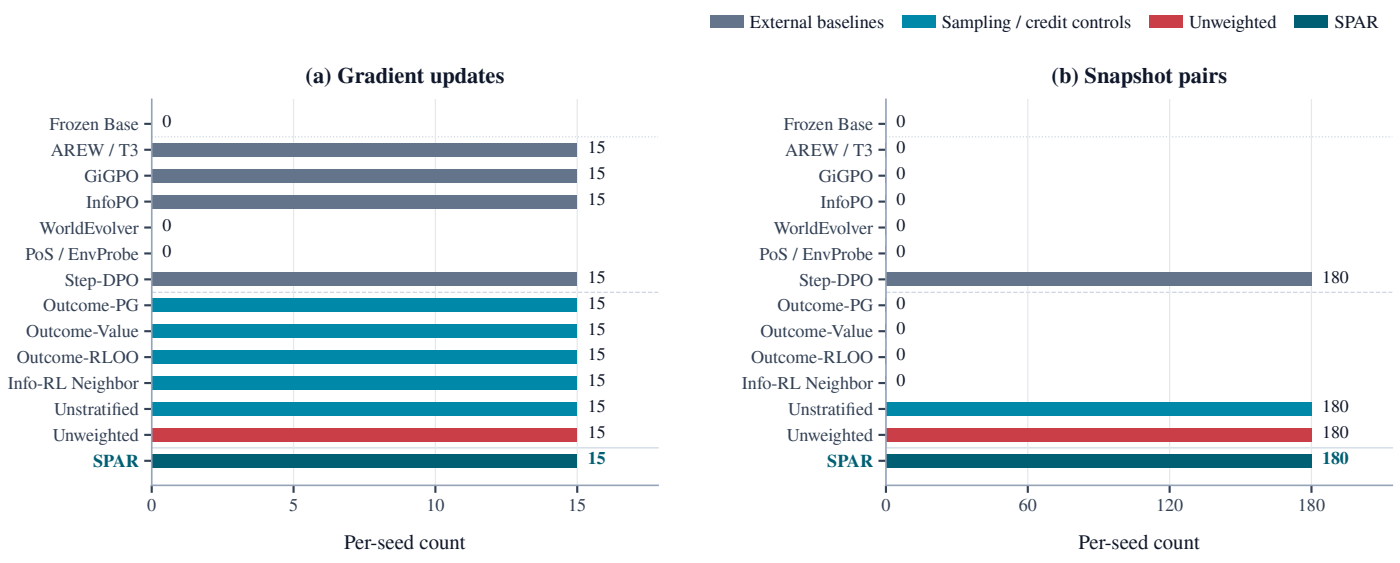
Method	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
	Model (s)	Grad. (s)	Restore (s)	Model (s)	Grad. (s)	Restore (s)	Model (s)	Grad. (s)	Restore (s)
Frozen Base	0	0	0	0	0	0	0	0	0
AREW / T3	150.2	26.84	0	89.4	26.54	0	277.1	4.40	0
GiGPO	149.0	27.66	0	89.0	65.24	0	274.2	4.41	0
InfoPO	147.9	29.41	0	90.6	56.07	0	265.7	5.23	0
WorldEvolver	124.3	0	0	81.5	0	0	245.4	0	0
PoS / EnvProbe	0	0	0	0	0	0	0	0	0
Step-DPO	124.4	35.91	0.007	85.1	21.75	0.003	261.1	6.17	0.010
Outcome-PG	152.0	22.52	0	89.0	35.16	0	277.9	5.26	0
Outcome-Value	150.3	31.89	0	88.6	69.34	0	271.0	3.81	0
Outcome-RLOO	150.5	31.49	0	88.4	35.66	0	277.8	3.81	0
Info-RL Neighbor	150.8	26.66	0	90.4	41.48	0	272.3	3.70	0
Unstratified	149.6	11.00	0.007	90.1	25.25	0	266.9	2.52	0.003
Unweighted	125.8	2.39	0.003	85.5	2.50	0.007	261.8	2.37	0.003
SPAR	**125.1**	**24.68**	**0**	**84.5**	**54.84**	**0.003**	**261.1**	**4.14**	**0.003**

(a) PlanBench-XL — Training interaction work and total wall-clock time per seed

Method	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
	Env. steps	Tokens (k)	Wall (s)	Env. steps	Tokens (k)	Wall (s)	Env. steps	Tokens (k)	Wall (s)
Frozen Base	0	0	0	0	0	0	0	0	0
AREW / T3	1289.7	582.5	5.73	1307.0	570.3	6.12	1280.0	579.1	5.42
GiGPO	1294.7	586.1	5.70	1299.7	566.4	5.96	1276.0	576.4	5.51
InfoPO	1288.0	581.6	6.49	1269.3	545.9	6.96	1265.7	571.0	6.31
WorldEvolver	1196.3	655.2	3.49	1193.3	632.5	3.73	1204.0	663.7	3.45
PoS / EnvProbe	0	0	0.54	0	0	0.68	0	0	0.59
Step-DPO	1303.3	593.7	7.97	1305.0	571.3	8.52	1309.7	603.2	7.85
Outcome-PG	1306.3	594.5	6.38	1302.3	568.4	6.98	1271.0	572.8	6.25
Outcome-Value	1298.3	589.8	5.85	1301.7	567.8	6.35	1269.7	572.2	5.48
Outcome-RLOO	1304.3	593.3	6.41	1303.3	568.6	7.00	1271.0	572.8	6.07
Info-RL Neighbor	1297.3	589.3	6.60	1287.3	558.1	7.14	1268.3	571.9	6.08
Unstratified	1292.3	584.5	5.89	1292.3	561.1	5.84	1274.3	578.2	5.43
Unweighted	1313.7	601.2	6.01	1314.0	578.0	6.01	1330.3	619.3	5.86
SPAR	**1322.3**	**607.3**	**6.12**	**1323.7**	**584.5**	**6.44**	**1329.0**	**618.3**	**6.04**

(b) PlanBench-XL — Model encoding, gradient-update, and snapshot-restore time per seed

Method	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
	Model (s)	Grad. (s)	Restore (s)	Model (s)	Grad. (s)	Restore (s)	Model (s)	Grad. (s)	Restore (s)
Frozen Base	0	0	0	0	0	0	0	0	0
AREW / T3	82.7	2.58	0	113.6	2.76	0	322.4	2.46	0
GiGPO	83.2	2.56	0	112.5	2.70	0	320.5	2.48	0
InfoPO	82.9	3.29	0	109.5	3.47	0	317.0	3.15	0
WorldEvolver	75.0	0	0	99.5	0	0	289.3	0	0
PoS / EnvProbe	0	0	0	0	0	0	0	0	0
Step-DPO	85.1	4.35	0	115.7	4.63	0.003	337.4	4.24	0
Outcome-PG	84.4	3.29	0	112.9	3.55	0	318.2	3.22	0
Outcome-Value	83.9	2.72	0	113.2	2.88	0	317.8	2.54	0
Outcome-RLOO	84.2	3.28	0	112.9	3.51	0	318.3	3.12	0
Info-RL Neighbor	83.9	3.41	0	111.2	3.64	0	318.2	3.10	0
Unstratified	83.0	2.30	0.003	113.4	2.23	0	322.0	2.07	0
Unweighted	86.3	2.21	0.007	117.2	2.20	0.003	347.1	2.18	0.003
SPAR	**87.1**	**2.24**	**0**	**118.6**	**2.36**	**0**	**346.5**	**2.17**	**0.010**


External baselines
Sampling / credit controls
Unweighted
SPAR
(a) Gradient updates
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
15
15
15
0
0
15
15
15
15
15
15
15
15
0
5
10
15
Per-seed count
(b) Snapshot pairs
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
0
0
0
0
0
180
0
0
0
0
180
180
180
0
60
120
180
Per-seed count